

## About Me

Innovative software engineering student with a strong foundation in building intelligent systems using LLMs, agentic AI, RAG, and Computer Vision. Demonstrated success delivering autonomous agents, production-grade full-stack platforms, and secure multi-role chatbots using modern Python and JavaScript frameworks. Holder of 6 patent filings, with 1 currently in the grant process.

## Education

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Vellore Institute of Technology

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*Aug 2023 – May 2028*

*Integrated M.Tech in Software Engineering*

- CGPA: 8.92 / 10.0

## Projects

## Role-Based Internal Chatbot with RBAC and RAG

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GitHub

- Developed an internal chatbot tailored for enterprise-grade RBAC, enhancing secure access to departmental data.
- Enabled role-specific responses from internal documents using RAG, FAISS vector search, and Groq LLM.
- Built using FastAPI for secure login, integrated with Streamlit UI for live interaction.
- **Tech Stack:** Python, FastAPI, Streamlit, FAISS, Sentence Transformers, Groq LLM

## Job Hunt AI Agent – Autonomous Job Search & Matching Platform

GitHub

- Built an autonomous agent that scrapes LinkedIn job postings via Playwright and bypasses basic bot detection to extract raw listings.
- Orchestrated a LangGraph pipeline feeding job descriptions to Gemini 3 Pro, generating a 0–100 match score, skill-gap analysis, and reasoning against the user’s resume.
- Implemented semantic deduplication using sentence-transformers embeddings and Supabase pgvector to automatically skip near-duplicate postings.
- **Tech Stack:** FastAPI, LangGraph, Gemini 3 Pro, Playwright, Supabase, Next.js, Tailwind/shadcn-ui

## KidTube – AI-Moderated, Parent-Controlled Video Platform

GitHub

- Architected a three-tier, whitelist-only video platform (parent dashboard, child mobile app, backend API) replacing recommendation-driven feeds with full parental curation.
- Built an AI thumbnail moderation pipeline using TensorFlow.js and nsfwjs with concurrent batch processing to screen channels before approval.
- Designed the backend with Chain-of-Responsibility auth middleware, factory-based input validation, and a Repository-pattern API client for maintainable, secure routes.
- **Tech Stack:** Next.js, Expo/React Native, Node.js/Express, Firebase Firestore, TensorFlow.js, JWT

## Achievements

- Filed 6 patents (1 currently in grant process); recognized as an “Achiever” by VIT (2025–26) for outstanding merit and national-level achievement.
- 2nd Place – Danfoss Innovator Award 2026.
- 1st Place – National-Level Hackathon, Sona College of Technology.
- 2nd Place – National-Level Hackathon, Sairam Engineering College.

## Skills

## Programming Languages:

- Python
- C++
- Java
- C
- JavaScript

## Technologies

- **AI/ML:** RAG, LangGraph, LangChain, Gemini, Groq LLM, LLaMA3, PyTorch, TensorFlow, sentence-transformers
- **Web/API:** FastAPI, Streamlit, Next.js, React.js, Node.js, Express, Expo/React Native
- **Data/Infra:** Firebase, Supabase (pgvector), PostgreSQL, Redis, FAISS, Docker
- **Tools:** OpenCV, NumPy, Playwright, YOLO (via PyTorch)

**Languages Known:** Tamil (Native), English (Fluent), Hindi (Basic Proficiency)